Practical No : 2

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Q.2) Write a C program for Binary Search.

```
#include <stdio.h>
int main()
{
   int arr[100], n, key, low, high, mid, i;
   printf("Enter the number of elements: ");
   scanf("%d", &n);
   printf("Enter the elements in sorted order:\n");
   for (i = 0; i < n; i++)
   {
      scanf("%d", &arr[i]);
   }
   printf("Enter the element to search: ");
   scanf("%d", &key);
   low = 0;
   high = n - 1;
   while (low <= high)
   {
      mid = (low + high) / 2;
      if (arr[mid] == key)
      {
         printf("Element found at position %d\n", mid + 1);
         return 0;
      }
      else if (arr[mid] < key)
      {
         low = mid + 1;
```

```
        }
        else
        {
            high = mid - 1;
        }
    }
    printf("Element not found.\n");
    return 0;
}
```

Output:

